THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Reproduction: How Life Continues** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Animals with external fertilisation lay a large number of eggs. Reason (R): Many eggs and sperms are lost in the water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A man's sperm count is low. It will most affect: 1

(a) chance of fertilisation **(b)** menstruation
(c) ovulation **(d)** puberty

3. A scion is the: 1

(a) flower **(b)** rooted part
(c) seed **(d)** twig of the desired plant

4. Coconut seeds are dispersed by: 1

(a) explosion **(b)** water **(c)** animals only **(d)** wind

5. The sperms are stored in the: 1

(a) vagina **(b)** ovary **(c)** uterus **(d)** epididymis

6. Assertion (A): A sperm determines the sex of the baby. Reason (R): Sperms carry either an X or a Y chromosome. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A couple has 2 boys. The chance that the third child will be a girl is: 1

(a) 1/2 **(b)** 1/4 **(c)** 1 **(d)** 1/8

8. The stock in grafting provides: 1

(a) flowers **(b)** fruits **(c)** seeds **(d)** the root system

9. **Assertion (A):** Self-pollination occurs when pollen from an anther reaches the stigma of a flower of a different plant. 1

Reason (R): Pollination is the transfer of pollen from the anther to the stigma.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Identical twins are always of the same sex. 1
Reason (R): Identical twins develop from a single fertilised egg that splits into two.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two reasons why twins from separate eggs are not identical. 2

12. Why is reproduction not essential for the survival of the individual but is essential for the species? 2

13. State why cross-pollination is considered better than self-pollination in an evolutionary sense. 2

14. A seed does not germinate in water-logged soil. Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A plant produced 10 flowers each with 5 ovules. If all ovules are fertilised, how many seeds? If 40 per cent of the seeds germinate, how many seedlings? 3

16. Describe the changes in the uterus during pregnancy and delivery with the hormones involved. 3

17. Discuss three ethical issues related to modern reproductive technologies and sex selection. 3

OR

Explain the adaptations of a seed for dispersal in coconut, maple, cotton, xanthium and balsam with the agent. 3

18. Describe the characteristics of a bisexual and a unisexual flower and give two examples of each. 3

OR

Compare the roles of the male and female parent in the fertilisation of an egg in a hen and in a frog. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rose plant in Bengaluru is grown from a stem cutting, while a mango tree is grown from seed. A flower of the mango has anthers that produce pollen and a pistil with stigma, style and ovary. 5
(i) Name the male and female reproductive parts of a flower.
(ii) What is pollination?
(iii) Name the method by which the rose is multiplied.
(iv) Give one advantage of growing plants by vegetative propagation instead of from seeds.

20. A farmer has a variety of mango with excellent fruit but the seeds give inferior trees. Explain why this happens and suggest the techniques with the steps for propagating the variety. 5

OR

Compare the life cycles of a flowering plant and a frog with respect to the gametes, the type of fertilisation, the development of the zygote and the young stage, and the way the species is dispersed. 5

END OF QUESTION PAPER — Total: 40 Marks